

Maximum Mark: 15

MidSem-2

Time: 60 minutes

1. (points) [2+3+3]

- (a) (points) [0.5+1.5] Starting with the classical Hamiltonian, find the time-independent Schrodinger equation for a Hydrogenic atom in the presence of a constant magnetic field (assume infinite nuclear mass approximation, show/state all steps/assumptions explicitly).
- (b) (points) [1+1] Assuming the external magnetic field across the z-direction, write the complete Schrodinger equation for a one-electron atom in a constant magnetic field including the full relativistic parts (*No derivation needed for the relativistic part. You may write the important functional part relevant for the Quantum system, while its rest of the factors in the expression can be taken as some function of the relevant variable; e.g. $(4\pi/r) \vec{S}_1 \cdot \vec{S}_2 = f(r) \vec{S}_1 \cdot \vec{S}_2$; stating only the later part will do (not exact constants and factors, but Functional form should be correct)*).
- (c) (points) [1+1+1] From the above, considering only the spin-orbit part, write the Schrodinger equation for weak and strong $\vec{B}$ field case (list unperturbed and perturbed H parts for each case separately). What decides the strong and weak in this case?

2. (points) [4+1+2]

- (a) For a two electron atom, write the complete set of spin wavefunctions and their respective eigen-values (S, M_S) explicitly and state the multiplicity as well as degeneracy (assuming electrons interaction as perturbative part).
- (b) (1 point) Write the complete wavefunction for a 2-electron atom (if needed, you may directly use result from above).
- (c) (2 points) For a 2-electron atom with one electron in the ground state and the other in the excited state ($n \geq 2$), express the 1-st order exchange correction under the independent particle approximation in the relevant spin (of the two electrons) form.

Useful Results

- $E_{nl,\pm}^{(1)} = J_{nl} \pm K_{nl}$
- $J = \int |\psi_{100}(\vec{r}_1)|^2 \frac{1}{r_{12}} |\psi_{nlm}(\vec{r}_2)|^2 d\vec{r}_1 d\vec{r}_2$
 $K = \int \psi_{100}^*(\vec{r}_1) \psi_{nlm}^*(\vec{r}_2) \frac{1}{r_{12}} \psi_{nlm}(\vec{r}_1) \psi_{100}(\vec{r}_2) d\vec{r}_1 d\vec{r}_2$